

1. For the following discrete-time system, where $y[n]$ and $x[n]$ are, respectively, the output and the input sequences, determine whether the system is (1) linear, (2) shift-invariant, (3) stable, and (4) causal. Note that detailed steps should be given in the solutions.

$$y[n] = 3(x[-n])^2 + 6$$

Solution:

- (1) Let $y_1[n]$ and $y_2[n]$ be the output of the discrete-time system, whose input are denoted by $x_1[n]$ and $x_2[n]$, respectively. Then

$$y_1[n] = 3(x_1[-n])^2 + 6$$

And
$$y_2[n] = 3(x_2[-n])^2 + 6$$

When the input is a linear combination of $x_1[n]$ and $x_2[n]$, i.e., $\alpha x_1[n] + \beta x_2[n]$, where α and β are both arbitrary real numbers, corresponding output $y_o[n]$ is given by

$$\begin{aligned} y_o[n] &= 3(\alpha x_1[-n] + \beta x_2[-n])^2 + 6 \\ &\neq \alpha y_1[n] + \beta y_2[n] \end{aligned}$$

The system is not linear.

- (2) Since

$$\begin{aligned} 3(x[-n-n_0])^2 + 6 &= 3(x[-(n+n_0)])^2 + 6 \\ &\neq y[n-n_0] \end{aligned}$$

The system is not shift-invariant.

- (3) If $|x[n]| \leq B_x$ (B_x is any positive bounded real number) for any integer value of n , then

$$|y[n]| = |3(x[-n])^2 + 6| \leq 3|x[-n]|^2 + 6 \leq 3B_x^2 + 6$$

Therefore, the discrete-time system is BIBO stable.

- (4) When $n < 0$, The system output $y[n]$ depends on the input $x[-n]$ where $-n > 0$. Therefore, the discrete-time system is not causal.
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2. A LTI discrete-time system is cascaded by following two systems shown as Fig1:

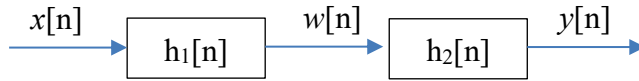


Fig 1

$h_1[n]$ is an ideal lowpass filter with $\omega_p = 0.5\pi$; The Linear Differential Equation of the

second system is $y[n] = w[n] + w[n-1]$ with $y[n] = 0, n < 0$;

1) Determine the frequency response $H_2(e^{j\omega})$ of the second system.

2) Determine the frequency response $H(e^{j\omega})$ of the whole system. Sketch its magnitude and phase response.

3) If $x[n] = \cos(0.6\pi n) + 2\delta[n-5] + 2$, Determine the output $y[n]$.

Solution

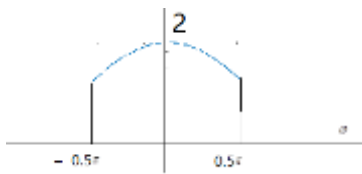
$$1) Y(e^{j\omega}) = W(e^{j\omega}) + e^{-j\omega}W(e^{-j\omega}), H_2(e^{j\omega}) = \frac{Y(e^{j\omega})}{W(e^{j\omega})} = 1 + e^{-j\omega}$$

$$2) H_1(e^{j\omega}) = \begin{cases} 1, & 0 \leq |\omega| \leq 0.5\pi \\ 0, & 0.5\pi \leq |\omega| \leq \pi \end{cases}, h_1[n] = \frac{\sin(0.5\pi n)}{\pi n}$$

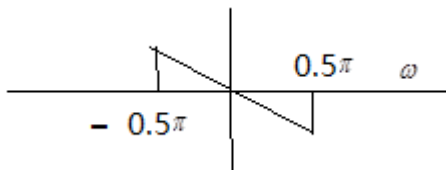
$$H(e^{-j\omega}) = H_1(e^{-j\omega})H_2(e^{-j\omega}) = \begin{cases} 1 + e^{-j\omega}, & 0 \leq |\omega| \leq 0.5\pi \\ 0, & 0.5\pi \leq |\omega| \leq \pi \end{cases}$$

Let $H(e^{-j\omega}) = |H(e^{j\omega})|e^{j\theta(\omega)}$, magnitude and phase response are:

$$\therefore |H(e^{j\omega})| = \begin{cases} |2\cos(\frac{\omega}{2})|, & 0 \leq |\omega| \leq 0.5\pi \\ 0, & 0.5\pi \leq |\omega| \leq \pi \end{cases}$$



$$\theta(\omega) = \begin{cases} -\frac{\omega}{2}, & 0 \leq |\omega| \leq 0.5\pi \\ 0, & 0.5\pi \leq |\omega| \leq \pi \end{cases}$$



(3)

$$x(e^{j\omega}) = \pi\{\delta(\omega + 0.6\pi) + \delta(\omega - 0.6\pi)\} + 2e^{-j5\omega} + 4\pi\delta(\omega) \quad |\omega| \leq \pi$$

$$W(e^{j\omega}) = X(e^{j\omega})H_1(e^{j\omega}) = 2e^{-j5\omega} + 4\pi\delta(\omega), |\omega| \leq 0.5\pi$$

$$W[n] = 2 \frac{\sin(0.5\pi(n-5))}{\pi(n-5)} + 2$$

$$y[n] = W[n] + W[n-1] = 2 \frac{\sin(0.5\pi(n-5))}{\pi(n-5)} + 2 \frac{\sin(0.5\pi(n-6))}{\pi(n-6)} + 4$$

3. Consider a discrete-time LTI system where the frequency response $H(e^{j\omega})$ is described by

$$H(e^{j\omega}) = \begin{cases} 1-2j, & 0 < \omega < \pi \\ 1+2j, & -\pi < \omega < 0 \end{cases}$$

(a) Is the impulse response of the system $h[n]$ real-valued?

(b) Calculate the following: $\sum_{n=-\infty}^{\infty} \left| \frac{1}{\sqrt{3}} h[n] \right|^2$

(c) Determine the response of the system to the input $x[n] = \cos(\omega_c n)$, where

$$0 < \omega_c < \frac{\pi}{2}.$$

Solution:

(a) The DTFT of $h[n]$ is conjugate symmetric. Therefore, $h[n]$ is real-valued.

(b) The calculation can be completed using Parseval's theorem. That is,

$$\sum_{n=-\infty}^{\infty} \left| \frac{1}{\sqrt{3}} h[n] \right|^2 = \frac{1}{3} \sum_{n=-\infty}^{\infty} |h[n]|^2 = \frac{1}{6\pi} \int_{-\pi}^{\pi} |H(e^{j\omega})|^2 d\omega = \frac{5}{3}$$

$$(c) \quad x[n] = \cos(\omega_c n) = \frac{1}{2} (e^{j\omega_c n} + e^{-j\omega_c n})$$

Considering that the output to $e^{j\omega_c n}$ is $e^{j\omega_c n} H(e^{j\omega_c})$ and the output to $e^{-j\omega_c n}$ is $e^{-j\omega_c n} H(e^{-j\omega_c})$, the response of the system to the input $x[n] = \cos(\omega_c n)$ is given by

$$y[n] = \cos(\omega_c n) + 2 \sin(\omega_c n)$$

4、 Let $x[n]$ 和 $X(e^{j\omega})$ represent a sequence and its Fourier transform, respectively. Determine, in terms of $X(e^{j\omega})$, the transform of $y_a[n]$, and $y_b[n]$. In each case, sketch $Y(e^{j\omega})$ for $X(e^{j\omega})$ as shown in Figure 9.

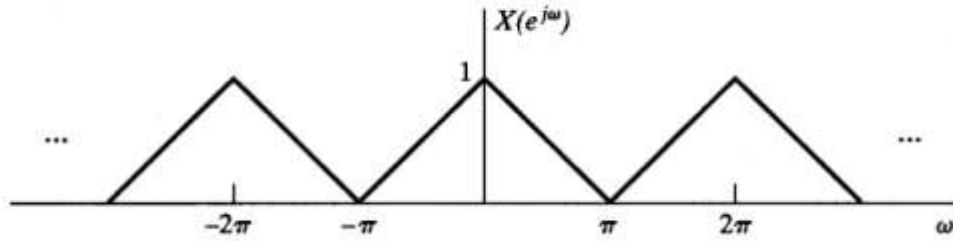


Figure 9

(a)

$$y_a[n] = \begin{cases} x[n], & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

(b)

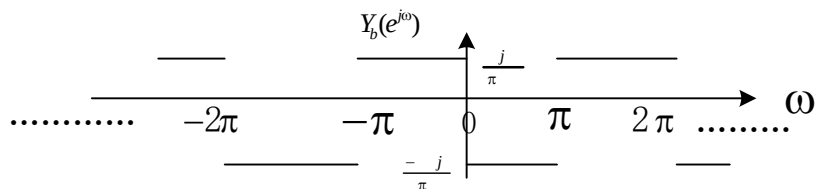
$$y_b[n] = nx[n]$$

Solution

(a) Note that $y_a[n] = \frac{1}{2} \{x[n] + (-1)^n x[n]\}$ and $-1 = e^{j\pi}$.

$$Y_a(e^{j\omega}) = \frac{1}{2} \{X(e^{j\omega}) + X(e^{j(\omega-\pi)})\}$$

(b) $Y_b(e^{j\omega}) = j \frac{d}{d\omega} X(e^{j\omega})$



5、About sampling:

(1) Please determine the smallest sampling rate for the following amplitude modulation (AM) signal $x_a(t)$, which satisfies the Nyquist rule.

$$x_a(t) = [1 + \cos(2\pi \times 100t)] \cos(2\pi \times 600t)$$

(2) Given $x_b(t) = \cos(2\pi ft)$, $f < 1\text{KHz}$

Sampling $x_b(t)$ with $f_s = 600\text{Hz}$, after recovery we obtain a frequency of 150Hz.

Sampling $x_b(t)$ with $f_s = 550\text{Hz}$, after recovery we obtain a frequency of 200Hz.

Please determine the value of f ?

Solution

$$\begin{aligned} (1) \quad x_a(t) &= \cos(1200\pi t) + \cos(200\pi t) \cos(1200\pi t) \\ &= \cos(1200\pi t) + \frac{1}{2} \cos(1000\pi t) + \frac{1}{2} \cos(1400\pi t) \end{aligned}$$

频率 f : 600Hz, 500Hz, 700Hz。

$$\text{采样频率 } f_{\text{sampling}} = 2 \times f_{\text{max}} = 1400\text{Hz}$$

so the smallest sampling rate is 1400Hz.

(2) we can know that $f > 300\text{Hz}$ from the question

$$\text{so } 300\text{Hz} < f < 1000\text{Hz}$$

$$\text{then } f = 150 + 600 = 750\text{Hz} \quad \text{or} \quad f = 600 - 150 = 450\text{Hz}$$

$$\text{and } f = 200 + 550 = 750\text{Hz} \quad \text{or} \quad f = 550 - 200 = 350\text{Hz}$$

In summary: $f = 750\text{Hz}$ The value of f is 750Hz.

6、A digital signal processing system is represented as figure 11.1. The frequency spectrum

$X_a(j\Omega)$ of an analog input signal $x_a(t)$ is given in figure 11.2. The analog signal is being

sampled at sampling interval $T = \frac{4\pi}{3\Omega_m}$. And pass the sampled sequence $g[n]$ through an ideal

lowpass filter $H(e^{j\omega})$ with cutoff frequency $\omega_c = \frac{\Omega_m T}{2}$, as shown in figure 11.3. Then put the

filter's output $y[n]$ into the ideal interpolator, an analog output $y_a(t)$ can be got. Please sketch:

(1) The frequency spectrum $G(e^{j\omega})$ of the sampled sequence $g[n]$.

(2) The frequency spectrum $Y(e^{j\omega})$ of the filter output $y[n]$.

(3) The frequency spectrum $Y_a(j\Omega)$ of the system output $y_a(t)$.

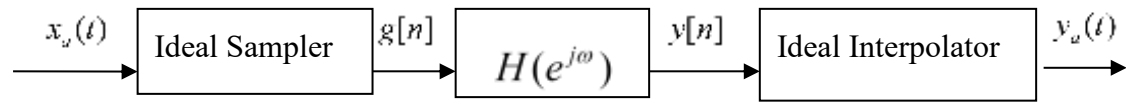


Figure 11.1 A digital signal processing system

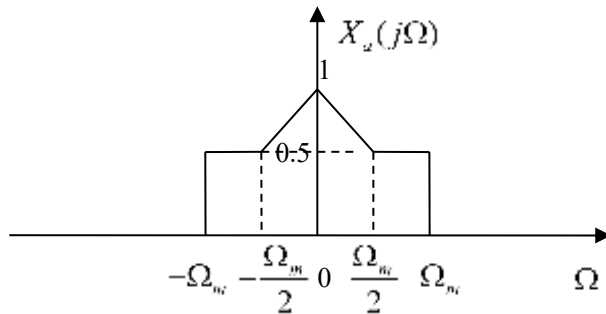


Figure 11.2 Spectrum of $x_a(t)$

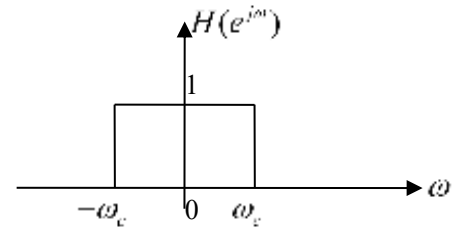
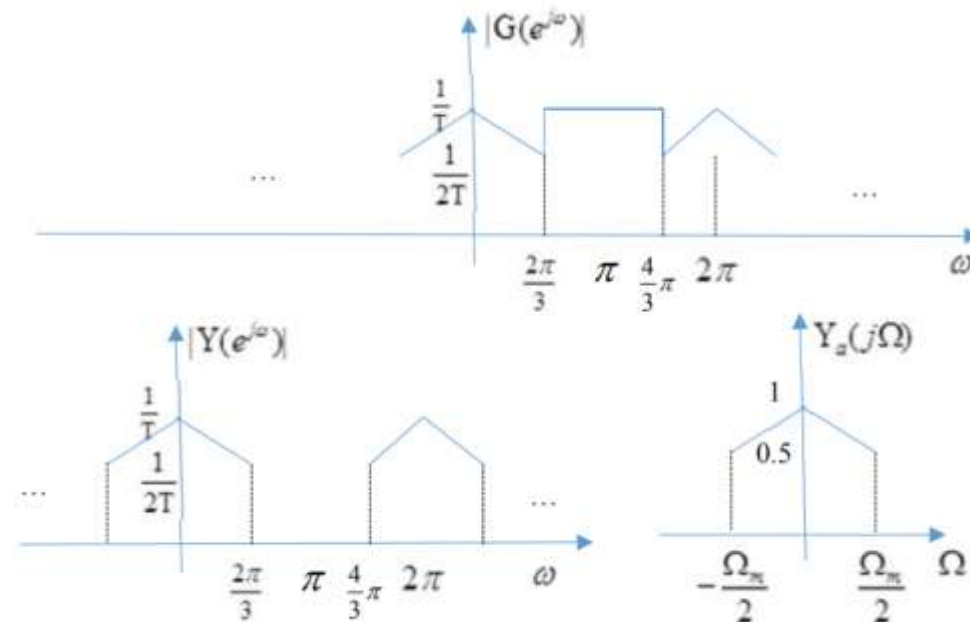


Figure 11.3 The spectrum of ideal lowpass filter

Solution:

$$\because T = \frac{4\pi}{3\Omega_m}, \therefore \Omega_s = \frac{2\pi}{T} = \frac{3\Omega_m}{2}, \therefore \omega_m = \Omega_m T = \frac{4}{3}\pi$$



7、A causal LTI discrete time system is described by the difference equation:

$$y[n] = 0.5y[n-1] + 0.25x[n] - x[n-1]$$

$x[n]$ and $y[n]$ are the input and output sequences of the system

- (1) Calculate the transfer function $H(z)$ of this system.
- (2) Give the zeros and poles of the system.
- (3) Find out a minimum phase system $H_{\min}(z)$ which has the same magnitude response as $H(z)$ (give your reason).

Solution

(1) $y[n] = 0.5y[n-1] + 0.25x[n] - x[n-1]$

$$Y(z) = 0.5z^{-1}Y(z) + 0.25X(z) - z^{-1}X(z) \quad H(z) = \frac{Y(z)}{X(z)} = \frac{0.25 - z^{-1}}{1 - 0.5z^{-1}}$$

(2) Poles: $1 - 0.5z^{-1} = 0, p_1 = 0.5$ Zeros: $0.25 - z^{-1} = 0, z_1 = 4$

(3) $H(z) = \frac{0.25 - z^{-1}}{1 - 0.5z^{-1}} = \frac{0.25 - z^{-1}}{1 - 0.25z^{-1}} \times \frac{1 - 0.25z^{-1}}{1 - 0.5z^{-1}} = A(z)H_{\min}(z)$

$A(z) = \frac{0.25 - z^{-1}}{1 - 0.25z^{-1}}$ is an All-pass system.

$H_{\min}(z) = \frac{1 - 0.25z^{-1}}{1 - 0.5z^{-1}}$ is a minimum phase system.

8、Known an 8-point real sequence $x[n]$ (shown as following Figure 2 with 8-point DFT $X[k]$).

(1) Sketch $\left(\frac{1}{8} \sum_{k=0}^7 X[k] e^{j(2\pi/8)kn} \right) \Big|_{n=9}$

(2) If $Y[k] = (-1)^k X[k]$, sketch $y[n]$ which is the 8-point IDFT of $Y[k]$.

(3) Determine the value of $X[4]$.

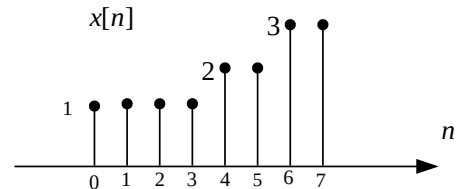


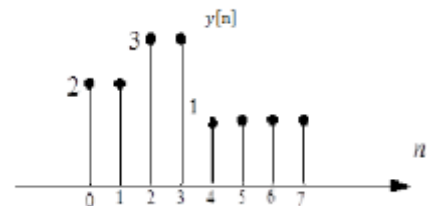
Fig 2

Solution (1) $X[k] = \sum_{n=0}^7 x[n] W_8^{kn}$, $x[n] = \frac{1}{8} \sum_{k=0}^7 X[k] W_8^{kn} = \frac{1}{8} \sum_{k=0}^7 X[k] e^{j(2\pi/8)kn}$

$x[9] = \left(\frac{1}{8} \sum_{k=0}^7 X[k] e^{j(2\pi/8)kn} \right) \Big|_{n=9}$ $x[\langle 9 \rangle_8] = x[1] = 1$

(2) $Y[k] = (-1)^k X[k]$

$y[n] = \frac{1}{8} \sum_{k=0}^7 (-1)^k X[k] W_8^{kn} = \frac{1}{8} \sum_{k=0}^7 X[k] W_8^{kn} W_8^{-k4}$
 $= x[\langle n - 4 \rangle_8]$



9、Let $H(z)$ be transfer function of a causal LTI system

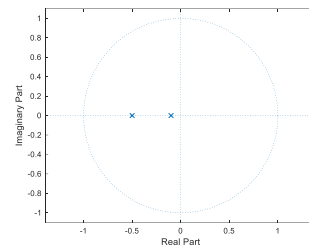
$$H(z) = \frac{z^{-2}}{1 + 0.6z^{-1} + 0.05z^{-2}}$$

- 1) Sketch the pole-zero plot of $H(z)$, and give it's ROC;
- 2) Give the difference equation of this discrete-time system;
- 3) Compute the $h[n]$ of this system;
- 4) Compute the frequency response of this system;
- 5) If $x[n]=u[n]$, give the stable response of this system.

Solution:

$$1) H(z) = \frac{1}{(z+0.5)(z+0.1)}, |z| > 0.5$$

6 分, ROC 2 分, 极点 2*2 分



$$2) y[n] + 0.6y[n-1] + 0.05y[n-2] = x[n-2] \quad 6 \text{ 分}$$

$$3) H(z) = \frac{1}{(z+0.5)(z+0.1)} = \frac{-2.5}{z+0.5} + \frac{2.5}{z+0.1} = \frac{-2.5z^{-1}}{1+0.5z^{-1}} + \frac{2.5z^{-1}}{1+0.1z^{-1}}, |z| > 0.5 \quad 6$$

$$h[n] = -2.5(-0.5)^{n-1}u[n-1] + 2.5(-0.1)^{n-1}u[n-1]$$

分

$$4) H(e^{j\omega}) = \frac{e^{-j2\omega}}{1 + 0.6e^{-j\omega} + 0.05e^{-j2\omega}} \quad 5 \text{ 分}$$

$$5) X(z) = \frac{1}{(1-z^{-1})}, |z| > 1$$

$$Y(z) = X(z)H(z) = \frac{-\frac{5}{6}}{z+0.5} + \frac{\frac{5}{22}}{z+0.1} + \frac{\frac{20}{33}}{z-1}, |z| > 1 \quad 2 \text{ 分}$$

$$y[n] = -\frac{5}{6}(-0.5)^{n-1}u[n-1] + \frac{5}{22}(-0.1)^{n-1}u[n-1] + \frac{20}{33}u[n-1] \quad 3 \text{ 分}$$

stable response:

$$y[n] \approx \frac{20}{33}, \quad n \geq 3 \quad 2 \text{ 分}$$

10

Consider a signal $x(t) = A \cos(2\pi f_1 t) + B \cos(2\pi f_2 t)$, $f_1 = 3k$ $f_2 = 4k$, Suppose

$x(t)$ is sampled with the $f_T = 20k$ to produce the discrete sequence $x[n]$.

Give the $x[n]$.

If $A=B=1$, **design a FIR filter** to block the f_1 , which should be suppressed by 60dB at least. You should use the **fixed window** function method and compute the **minimum length of window**. Give the mathematical expression of your FIR filter.

Solution.

$$x[n] = A \cos(0.3\pi n) + B \cos(0.4\pi n)$$

$$\omega_p = 0.4\pi \quad \omega_s = 0.3\pi \quad \alpha_s = 60dB$$

Choose Blackman window, $5.56\pi / M = 0.1\pi \Rightarrow M \approx 56$ $\omega_c = 0.35\pi$

$$w[n] = \left[0.42 + 0.5 \cos\left(\frac{2\pi(n-M)}{2M+1}\right) + 0.08 \cos\left(\frac{4\pi(n-M)}{2M+1}\right) \right], 0 \leq n \leq 2M$$

$$\hat{h}_{HP}[n] = \begin{cases} -\frac{\sin(\omega_c(n-M))}{\pi(n-M)}, & 0 \leq n \leq 2M, \text{ except } n = M \\ 1 - \frac{\omega_c}{\pi}, & n = M \text{ only} \end{cases}$$

FIR filter is: $h[n] = \hat{h}_{HP}[n] \cdot w[n], 0 \leq n \leq 2M$

11

Use bilinear transformation method to design a low-pass Butterworth digital filter with the

following specifications: the passband edge frequency $\omega_p = \frac{\pi}{3}$, the stopband edge

frequency $\omega_s = \frac{2\pi}{3}$, the maximum passband attenuation $\alpha_p = 0.6dB$ and the minimum

stopband attenuation $\alpha_s = -10dB$.

1) Calculate the prewarp frequencies Ω_p and Ω_s .

2) Determine minimum order N of the Butterworth filter

3) Determine the 3-dB cutoff frequency Ω_c when the **stopband** specification is exactly matched.

4) Use the results calculated in steps 2) and 3) to determine the analog prototype filter $H(s)$.

5) Use bilinear transformation to convert $H(s)$ into the desired digital filter $H(z)$

6) Draw a canonic structure realization of $H(z)$.

$$(1) \Omega_p = \tan\left(\frac{\omega_p}{2}\right) = \tan\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{3} \quad (3 \text{ 分})$$

$$\Omega_s = \tan\left(\frac{\omega_s}{2}\right) = \tan\left(\frac{\pi}{3}\right) = \sqrt{3} \quad (3 \text{ 分})$$

$$(2) \alpha_p = 20 \log_{10} \sqrt{1 + \varepsilon^2} \Rightarrow \varepsilon^2 = 10^{\frac{0.6}{10}} - 1 = 0.148$$

$$\alpha_s = 20 \log_{10} \frac{1}{A} \Rightarrow A = 10^{\frac{1}{2}} \Rightarrow A^2 = 10$$

$$\text{所以 } k = \frac{\Omega_p}{\Omega_s} = \frac{\frac{\sqrt{3}}{3}}{\sqrt{3}} = \frac{1}{3} \quad k_1 = \frac{\varepsilon}{\sqrt{A^2 - 1}} = \frac{\sqrt{0.148}}{\sqrt{9}} = \frac{\sqrt{0.148}}{3}$$

$$N = \frac{\log_{10}\left(\frac{1}{k_1}\right)}{\log_{10}\left(\frac{1}{k}\right)} = \frac{\log_{10}\left(\frac{3}{\sqrt{0.148}}\right)}{\log_{10} 3} = 1.8695 \quad \text{所以 } N = 2 \quad (\text{共 } 6 \text{ 分})$$

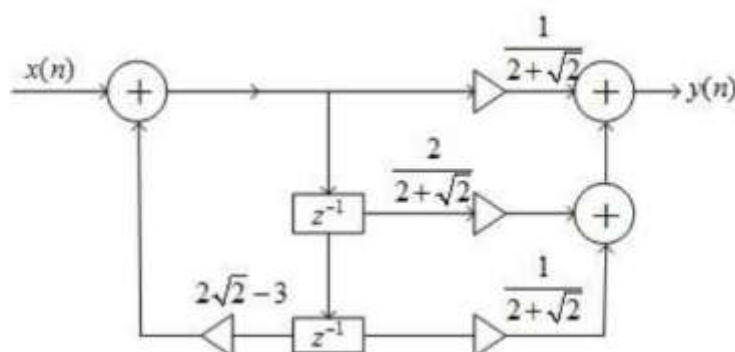
$$(3) \frac{1}{\left(1 + \frac{\Omega_s}{\Omega_c}\right)^{2N}} = \frac{1}{A^2} \Rightarrow \Omega_c = \Omega_s (A^2 - 1)^{\frac{1}{2N}} = \sqrt{3} \times (10 - 1)^{\frac{1}{4}} = 1 \quad (3 \text{ 分})$$

$$(4) H(s) = \frac{1}{s^2 + \sqrt{2}s + 1} \quad (3 \text{ 分})$$

$$(5) H(z) = H(s) \Big|_{s=\frac{1-z^{-1}}{1+z^{-1}}} \Rightarrow H(z) = \frac{1}{\left(\frac{1-z^{-1}}{1+z^{-1}}\right)^2 + \sqrt{2} \frac{1-z^{-1}}{1+z^{-1}} + 1}$$

$$= \frac{1 + 2z^{-1} + z^{-2}}{(2 + \sqrt{2}) + (2 - \sqrt{2})z^{-2}} \quad (3 \text{ 分})$$

$$(6) \quad (3 \text{ 分})$$



12、 The pole-zero plots in Figure 15 describe six different causal LTI systems.

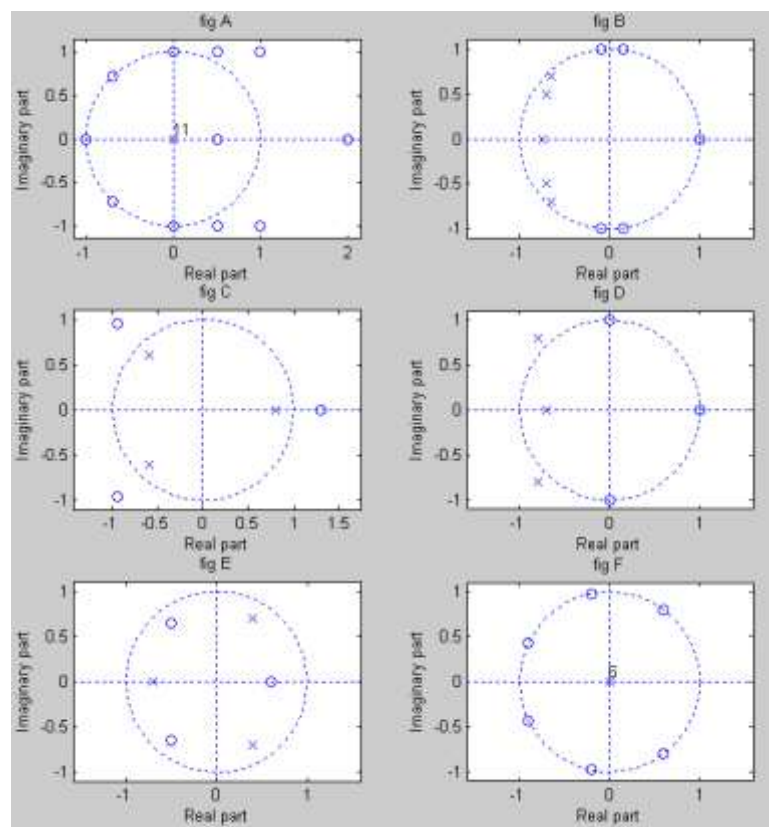


Figure15 Pole-zero plots of six different causal LTI system

Answer the following questions about systems having the above pole-zero plots. In each case, an acceptable answer could be none or all.

- 1) Which systems are IIR system? Why?
- 2) Which systems are FIR system? Why?
- 3) Which systems are stable system? Why?
- 4) Which systems are minimum-phase system? Why?
- 5) Which systems are linear-phase system? Why?
- 6) Which systems have $|H(e^{j\omega})| = \text{constant}$ for all ω ? Why?
- 7) Which systems has the shortest impulse response? Why?
- 8) Which systems have lowpass frequency responses? Why?

13、

1 . Use **fixed** windows to design a lowpass FIR filter with a linear phase. When the following signal $x[n]$ pass this filter, the higher frequency should be suppressed by at least 50dB. Give the specifications of this filter, choose a window and explain your reasons, and give the impulse response of this filter.

$$x[n] = \cos\left(\frac{\pi n}{6}\right) + \frac{3}{4} \cos\left(\frac{\pi n}{4}\right)$$

2 .If use a rectangular window with length-N to multiply $x[n]$, and then do N-point DFT analysis.

What is the smallest N, which can clearly resolve(分辨) these two sinusoids of $x[n]$ in magnitude of DFT? Sketch the magnitude of DFT.

1) choosing Hamming window for 50dB suppression.

$$\omega_p = \frac{\pi}{6}, \quad \omega_s = \frac{\pi}{4}, \quad \Delta\omega = \omega_s - \omega_p = \frac{\pi}{12}, \quad \omega_c = \frac{\omega_p + \omega_s}{2} = \frac{5\pi}{24}$$

$$\text{From table 10.2, } \frac{3.32\pi}{M} \leq \frac{\pi}{12} \Rightarrow M \geq 39.84, \text{ let } M = 40$$

$$\text{Ideal LPF } h_d[n] = \frac{\sin(\omega_c n)}{\pi n} = \frac{\sin\left(\frac{5\pi n}{24}\right)}{\pi n}$$

$$w[n] = 0.54 + 0.46 \cos\left(\frac{\pi n}{40}\right), \quad -40 \leq n \leq 40$$

So, the causal LPF is

$$h_i[n] = h_d[n-M]w[n-M] = \frac{\sin\left(\frac{5\pi(n-40)}{24}\right)}{\pi(n-40)} [0.54 + 0.46 \cos\left(\frac{\pi(n-40)}{40}\right)], \quad 0 \leq n \leq 80$$

N-point DFT has resolution $\frac{2\pi}{N}$. When $x[n]$ is analyzed with N-point DFT, it is

$$\text{required } \frac{2\pi}{N} \leq \Delta\omega = \frac{\pi}{12} \Rightarrow N \geq 24$$

We can roughly sketch the magnitude of 24-point DFT as following



14、Design a linear-phase FIR system $H(e^{j\omega})$ by windowing. The specifications are given as follow

$$\begin{cases} 0.9 \leq |H(e^{j\omega})| \leq 1, & |\omega| \leq 0.1\pi \\ |H(e^{j\omega})| \leq 0.01, & 0.5\pi \leq |\omega| < \pi \end{cases}$$

- (1) Find the **possible choices** for the type of fixed window
- (2) If the **Hamming** window is employed, determine the **minimum N** and **group delay** of the design for **type II** linear-phase FIR, where M is defined as follow

$$\text{Window}[n] = \begin{cases} \text{Non-zero}, & 0 \leq n \leq N \\ 0, & \text{others} \end{cases}$$

- (3) Based on question2, if we want to construct the filter by **Linear-phase direct form**, how many **multipliers, delayers and adders** shall we need?
- (4) If we want to design an IIR filter by **bilinear transform** to satisfy $|H(e^{j\omega})|$, indicate the specifications of analog prototype filter(模拟原型滤波器). (Note: sampling frequency T=2)
- (5) Compare the **advantages** of IIR and FIR filter.

A: (1)窗类型决定了滤波器设计的逼近误差

通带逼近误差 $\delta_1 < 0.1$ ，阻带逼近误差 $\delta_2 < 0.01$

$$\rightarrow \delta < \min(\delta_1, \delta_2) = 0.01 = -40dB$$

→Hanning, Hamming 和 Blackman 均可使用.

- (2) 采用 Hamming 窗，其过渡带 $\Delta\omega = 3.32\pi/M$

滤波器设计要求过渡带 $\Delta\omega = 0.5\pi - 0.1\pi = 0.4\pi$

$M > 8.3$, 由于采用 TypeII 类型，要求 N 取奇数，N=17, 其群延迟=8.5

- (3) TypeII 线性相位结构需要运算资源：

加法器数量：17 个；延迟单元数量：17 个；乘法器数量：9 个

- (4) 双线性变换，模拟原型滤波器设计指标为

$$\begin{cases} 0.9 \leq |H(j\Omega)| \leq 1, & |\Omega| \leq \tan(0.05\pi) \\ |H(j\Omega)| \leq 0.01, & |\Omega| \geq \tan(0.25\pi) \end{cases}$$

- (5) IIR 滤波器与 FIR 滤波器比较

- ① 相同的技术指标下，IIR 滤波器的阶数比 FIR 滤波器的少。
- ② FIR 滤波器可得到严格线性相位，而 IIR 滤波器则难以做到。
- ③ IIR 滤波器只有当所有极点都在单位圆内时滤波器才是稳定的，FIR 滤波器均稳定。
- ④ 相同阶数下，FIR 滤波器由于可采用 FFT 来实现，因此运算速度更快
- ⑤ IIR 滤波器可利用 C-T 滤波器现成的数据和表格来设计，计算工作量较小。

Q2 Consider the discrete-time LTI system with real impulse response $h[n]$ and its frequency response as following form: $H(e^{j\omega}) = A(\omega)e^{j\phi(\omega)}$,

where $A(\omega) \geq 0$ and $\phi(\omega)$ are real valued functions. Let $x[n]$ and $y[n]$ denote the input and output of the system respectively.

1) If the input $x_1[n] = 2$, evaluate the output $y_1[n]$; [5]

2) If the input $x_2[n] = \frac{1}{2}\cos(\omega_0 n) + \frac{1}{2}(-1)^n$, evaluate the output $y_2[n]$; [8]

3) Suppose $h[n] = (\delta[n] + 2\delta[n-1] + \delta[n-2]) / 4$. Determine the functions $A(\omega)$ and $\phi(\omega)$ on the interval $0 < \omega < \pi$; [7]

4) Give the $y_1[n]$ and $y_2[n]$ based on the $h[n]$ in 3). [5]

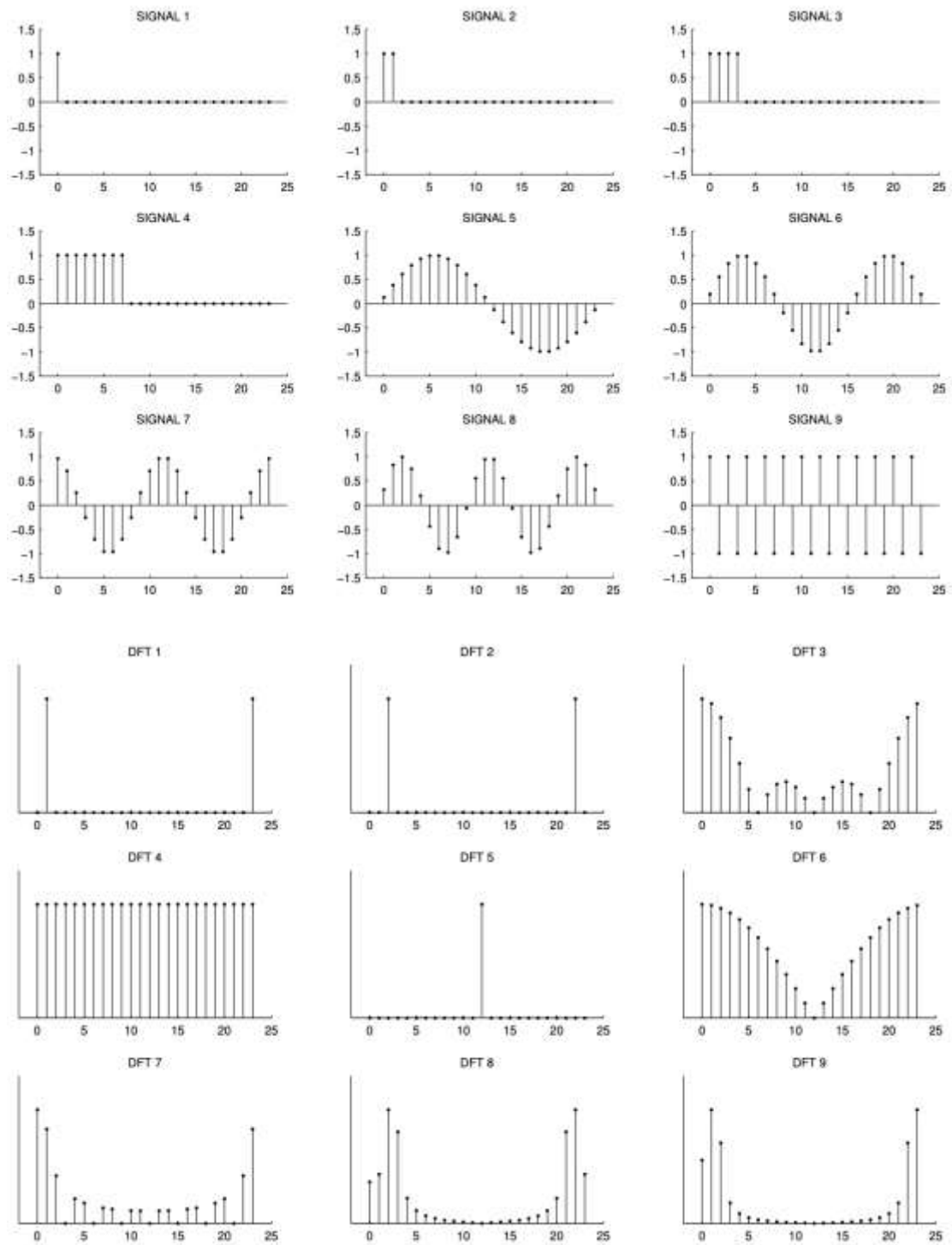
A: 1) $x_1[n] = 2 \cdot e^{j0} \Rightarrow y_1[n] = 2A(0)e^{j\phi(0)}$ $\because h[n]$ is real, $\therefore \phi(0) = 0$.
 Therefore, $y_1[n] = 2A(0)$ (3 points)
 and $H(e^{j\omega}) = H^*(e^{-j\omega})$ (2 points)

2) $x_2[n] = \frac{1}{4}(e^{j\omega_0 n} + e^{-j\omega_0 n}) + \frac{1}{2}e^{j\pi n}$ (3 points)
 Therefore, $y_2[n] = \frac{1}{4}A(\omega_0)e^{j\phi(\omega_0)}e^{j\omega_0 n} + \frac{1}{4}A(-\omega_0)e^{j\phi(-\omega_0)}e^{-j\omega_0 n}$
 $+ \frac{1}{2}A(\pi)e^{j\phi(\pi)}e^{j\pi n}$
 $= \frac{1}{2}A(\omega_0)\cos(\omega_0 n + \phi(\omega_0)) + \frac{1}{4}A(\pi)e^{j\phi(\pi)}(-1)^n$ (5 points)

3) $H(e^{j\omega}) = \frac{1}{4}(1 + 2e^{-j\omega} + e^{-j2\omega}) = \frac{1}{4}e^{-j\omega}(e^{j\omega} + 2 + e^{-j\omega})$
 $= \frac{1}{2}e^{-j\omega}(1 + \cos\omega)$ (3 points)
 $\therefore A(\omega) = \frac{1}{2}(1 + \cos\omega)$, $\phi(\omega) = -\omega$ (4 points)

4) From 3),
 $y_1[n] = 2 \cdot \frac{1}{2}(1 + \cos(0))e^{j0} = 2$ (2 points)
 $y_2[n] = \frac{1}{2} \cdot \frac{1}{2}(1 + \cos\omega_0)\cos(\omega_0 n - \omega_0) + \frac{1}{2} \cdot \frac{1}{2}(1 + \cos\pi)e^{j\pi} \cdot (-1)^n$
 $= \frac{1}{4}(1 + \cos\omega_0)\cos(\omega_0(n-1))$ (3 points)

Q4



1-DFT4 2-DFT6 3-DFT3 4-DFT7 5-DFT1 6-DFT9 7-DFT2 8-DFT8 9-DFT5